

Python Apartment Finder

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Objective

The Objective of this project is to provide an apartment hunting tool that aggregates rental listings from multiple sources that combines them with neighborhood and commute data to identify the best apartment values. This project will collect data and perform analysis to answer questions like: Which apartments offer the best price per square foot? Which neighborhoods provide the best value based on price, walkability, and location? This project will use web scraping, APIs, and data analysis.

Data Sources

Source 1: <https://www.craigslist.com/>

The data to be collected is the price, number of bedrooms/bathrooms, square footage, address, city, listing url, Listings will be scraped using Python requests and BeautifulSoup to extract structured apartment listing data.

Source 2: Walkability and Neighborhood Data (API): <https://www.walkscore.com/professional/api.php>

The data to be collected will be the walk score, transit score, and bike score. The method being used will be API calls based on apartment locations latitude/longitude to retrieve neighborhood scores.

Source 3: Commute Time Data (API)

OpenRouteService API: <https://api.openrouteservice.org/>

Data collected will be commute time to specific destinations and the distance.

Method:

The program will use coordinates from apartment listings to compute commute time to a fixed destination, I will be using the apartment location to my work (El Segundo), and school (USC Campus) to be able to get different commute times and add that data to the rest.

How the Sources Combine:

The apartment listing dataset will serve as the base dataset. Addresses from apartment listings will be converted to latitude and longitude coordinates. These coordinates will be used to query the Walk Score API to retrieve neighborhood data and the commute API to retrieve travel time information.

Final combined dataset example:

Address	Price	Beds	Sqft	Price/Sqft	WalkScore	Commute Time	City
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Questions This Project Will Answer

- Which neighborhoods offer the lowest price per square foot? (Cheapest)
- Which neighborhoods provide the best value for the price? (Best Value)
- Which neighborhoods have the best value for commute time?
- Which neighborhoods have the best value for walk score?

Analysis and Visualization

The integrated data will be analyzed to see deciding factors for renters. These can include average rent by city, price per square foot, and correlations between rent and commute time. Visualizations will include scatter plots of rent versus commute time, bar charts of average rent by neighborhood, and rankings of top-value apartments. These visualizations will help identify trends and provide help for apartment selection.